

Professional Data Portfolio Strategy: Business Intelligence Analyst

This document outlines the strategic roadmap for developing a high-impact professional portfolio designed to demonstrate end-to-end data engineering and business intelligence capabilities. The primary focus is on "Event-Driven Analytics," showcasing technical proficiency in SQL, Python, and Power BI.

Project 1: Trump Social Media & Market Correlation (Primary Project)

This project serves as the anchor for the portfolio, demonstrating the ability to handle unstructured text data, perform sentiment analysis, and correlate it with structured financial time-series data.

Technical Workflow

- **Data Acquisition:** Utilize the comprehensive "Trump Tweets 2009-2025" dataset (Data Driven Decision Lab), which includes columns: id, date, platform, handle, text, favorite_count, repost_count, quote_flag, repost_flag, and deleted_flag.
- **Data Engineering (Python):**
 - Clean the data using flags (deleted_flag, quote_flag).
 - Perform Natural Language Processing (NLP) using FinBERT or VADER to generate a sentiment_score (-1.0 to 1.0) for each tweet.
- **Data Modeling (SQL):**
 - Architect a Star Schema in PostgreSQL.
 - **Fact Table:** Fact_Tweets (TweetID, Date, SentimentScore, Engagement).
 - **Dimension Tables:** Dim_Date (Time-series metadata) and Dim_Market (S&P 500 or BTC-USD price data).
 - **SQL Logic:** Implement Window Functions for rolling averages and LEFT JOINs to manage non-trading days.
- **Visualization (Power BI):**
 - Interactive Dashboard featuring sentiment vs. volatility trend lines.
 - Platform-specific slicer (Twitter vs. Truth Social).
 - Python visual for correlation heatmaps.

Strategic Narrative

Framing: "I developed a pipeline to analyze 16 years of social media discourse, utilizing Python for NLP and SQL for relational modeling, to identify statistically significant correlations between rhetorical shifts and financial market volatility."

Additional Portfolio Projects

Project	Focus Area	Key Skills
LeBron Career	Performance Analytics	DAX, Rolling Averages, Complex Modeling
Deloitte	Firm Benchmarking	Qualitative-to-Quantitative Mapping, Business Case Modeling
DHL	Supply Chain Optimization	Operational Metrics, Throughput, Lead-time Analysis

Portfolio Deployment Strategy

- **Platform:** Next.js on Vercel for a professional "Digital Resume" aesthetic.
- **GitHub:** A dedicated repository per project containing README.md (Business Problem, Methodology, Key Findings) and an /sql_queries folder showcasing complex joins and window functions.
- **Action:** Link the live Power BI dashboard and GitHub code within each project card on the portfolio site.